

Patrick Engelbert

Waco, TX • (785) 418-1614 • patrickengelbert2@gmail.com

Portfolio: patrickengelbert.com • LinkedIn • GitHub

Robotics & Industrial Controls Student | Expected Graduation: Dec 2026 | PLC Programming | FANUC & Universal Robotics

PROFILE

Robotics & Industrial Controls student at Texas State Technical College, graduating Dec 2026, with hands-on experience in Studio 5000 PLC programming, FANUC robot programming, industrial circuits, sensors, motor controls, and technical documentation. Prior production software engineering experience strengthens troubleshooting, documentation, and disciplined execution.

TECHNICAL SKILLS

Controls & Automation: Rockwell Studio 5000, ladder logic, relays, start/stop latching, timers, sequencing, troubleshooting, industrial I/O

Robotics & Motion: FANUC 200i, teach pendant operation, position registers, offset-based motion, pick-and-place sequencing, auto-mode validation

Electrical & Documentation: AC/DC circuits, sensors, motor controls, electro-mechanical devices, schematic reading, AutoCAD, parts lists, sequence of operation

Programming & Tools: JavaScript, TypeScript, Python, Git/GitHub, Angular, React, and more

SELECTED AUTOMATION PROJECTS

Conveyor Control Circuit | Start/Stop Latch, Inductive Sensor, Documentation

- Designed a conveyor control circuit with start/stop latch logic and an inductive-sensor interlock that automatically pauses movement when a target is detected and resumes when clear.
- Created maintenance-ready documentation including project purpose, circuit diagram, complete parts list, and sequence of operation.

PLC Arcade Machine | Rockwell Studio 5000

- Programmed an 8-light reflex game with operator pushbutton input, timer-based sequencing, and six selectable difficulty modes: Easy, Normal, Hard, and Turbo variants of each.
- Added randomized turbo-speed logic after each pass to demonstrate timing variation, state control, and responsive operator interaction.

FANUC 200i Pick-and-Place Program | Offset-Based Robot Programming

- Reduced teach time by manually teaching one reference point and calculating the remaining eight slot positions in-program using offsets and position registers.
- Validated repeatability by running the 9-slot pick-and-place sequence overnight in auto mode with no crashes or faults.

EDUCATION

Robotics & Industrial Controls, Texas State Technical College

Expected Dec 2026

- Coursework: Robotics, DC Circuits, AutoCAD, AC Circuits, Motor Controls, Electro-Mechanical Devices, Robot Programming, Sensors, and PLC Programming.

Software Engineering Program, Thinkful

2021-2022

Associate of Science and Associate of Arts, Neosho County Community College

2020

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer, Citizens Bank

Mar 2022 - Nov 2023

Johnston, RI

- Built and supported customer-facing web components using Angular, TypeScript, and Node.js within an Agile development team.
- Worked in production environments requiring careful testing, secure data handling, change discipline, and collaboration across teams; strengthened troubleshooting, documentation, and dependable execution under real delivery constraints.

ADDITIONAL EXPERIENCE

Pharmacy Technician, Cosentino's Price Chopper (2021); **After School Teacher**, RFTS (2019-2020); **Sales Associate**, Walmart SuperCenter (2015-2019); **Bank Teller**, Kansas State Bank (2014-2015)

- Earlier roles strengthened accuracy, customer interaction, organization, communication, and dependable execution in fast-paced environments.